

destroyed over 60 per cent of the Harrier fleet. Yet at least one crucial factor unifies most recent crashes: all of them occurred while the aircraft were making their landing approach suggesting a loss of control and loss of power in the final stages of landing. Clearly, an engine related issue in an aircraft that is already 25 years old. "The navy conscientiously maintains its aircraft but perhaps there are some components deep inside which are not apparent at regular inspections," says former navy chief Admiral (retired) Arun Prakash, who flew in the first Harriers in 1983.

The spate of crashes came to the attention of Right to Information (RTI)

fly in the next six months. This upgrade will deliver to the navy a formidable interceptor. Its new Elta EL/M-2032 radar can spot the enemy from over 100 km away and its Derby Beyond Visual Range (BVR) air-to-air missiles can engage them at 60 km. Unfortunately, at the current crash rate of nearly two aircraft every year it seems doubtful that there will be any Harriers left to last through another decade the navy wants them to be in service.

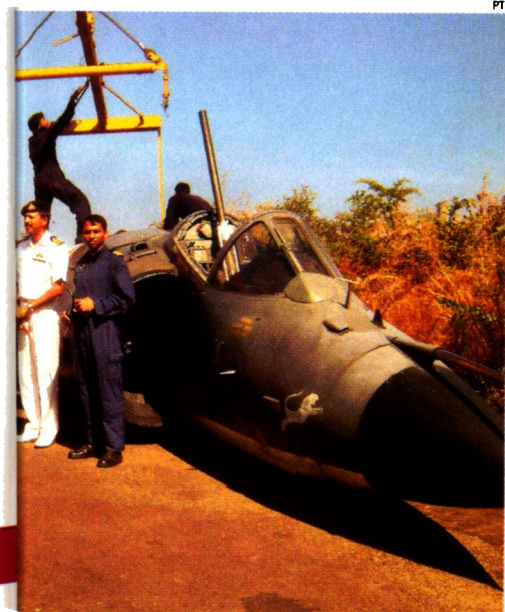
Naval officials blame the UK, the original manufacturer of the aircraft for stalling urgently needed upgrades to the Harriers, first proposed in the mid-1990s. The V/STOL aircraft, which

Two years ago, the then defence minister Pranab Mukherjee announced that HAL would upgrade Sea Harriers. The limited upgrade of the 13 aircraft would give them Israeli Elta radars and Derby BVR missiles. The aircraft are to be delivered by 2009 and will serve the navy for another decade. With the navy deciding to extend the life of the *Viraat* by another decade and the Harrier already out of production in the world, analysts feel it has no option but to purchase second-hand aircraft from the UK.

The navy wanted to purchase the eight remaining Sea Harrier FA2 aircraft from the Royal Navy last year as crash replacements, but a waffling British government said it would supply them only after stripping them of radars, avionics and missile, a proposal that was rejected by the navy. During a visit to New Delhi last year, Britain's First Sea Lord Admiral Sir Jonathon Band agreed to supply the navy the FA2s without the radars and avionics so they could be cannibalised for spare parts, but the UK subsequently backed out of this offer. Now, it turns out, there may be no more Sea Harriers left to buy.

With the Royal Navy having withdrawn its Sea Harriers from service in March last year (Royal Navy carriers operate the ground-attack variants of the jets), the Indian Navy now enjoys the uncomfortable distinction of being the world's last operator of the aircraft which first flew in 1960. With the crashes having created an unacceptable void in its capabilities, the navy may have no option but to scout the world market for second-hand Harrier cousins like the AV-8B Harrier-II operated by the US Marine Corps and the "Matador" operated by the Spanish Navy. Here too there are problems. Unlike the British Aerospace-built Harriers which the Indian Navy operates, the AV-8Bs are built by the US McDonnell Douglas. According to naval aviators, these will bring their own set of maintenance hassles.

In the absence of any alternatives, the navy has no option but to thrust the burden of defending its growing fleet of surface warships on the ageing shoulders of its shrinking fleet of Harriers. ■



CRASH COURSE

WHY WE NEED THE HARRIER: For fleet air defence. Embarked on the aircraft carrier *Viraat*, it protects the fleet from enemy aircraft, warships and submarines.

DELAY IN VIKRAMADITYA REFT: Five-year delay in the refit of the carrier in Russia means the navy will have only the Harriers to protect its fleet.

NUMBER OF CRASHES: In the past 20 years, the Harrier fleet is down from about 30 aircraft to 13 or less than the minimum requirement of a squadron of 18 aircraft.

DELAYS BY UK GOVT: Refused to upgrade Indian Sea Harriers with advanced radars and missiles since the mid-1990s. Did not give last batch of its aircraft for spare parts.

WORRISOME TREND: Crashed Sea Harrier at Goa's Dabolim airport

activists Hari Kumar P. and Frederick Noronha who petitioned and successfully obtained a list of the crashes from the navy. "The crashes have been occurring with worrying frequency in Goa and seemed to have escaped any national debate," says Noronha.

The navy says upgrades will reduce the risk of further accidents. "We have begun an upgradation of sensors, avionics and airframe of the Sea Harriers to improve the pilot's situational awareness and lessen his load," says naval spokesperson Commander Nirad Sinha. The upgrade of the Harriers is now being carried out by Hindustan Aeronautics Ltd (HAL) in Bangalore and the first Limited Upgrade Sea Harrier (LUSH) jet is set to

can take off from half the runway length required by normal fighter jets, were built by British Aerospace and first inducted from the UK in 1983 to operate off the British-built *Viraat* and the since decommissioned *INS Vikrant*. Their upgrades to the aircraft, pending since the mid-1990s were delayed by a series of flip-flops by the British government and their refusal to part with the advanced Blue Vixen radar and US-supplied AMRAAM BVR missiles. This led to the shelving of the first upgrade. With the euphoria generated by the acquisition of *Vikramaditya* and its air wing of 18 MiG-29K fighter aircraft in 2004, the navy decided to abandon the Harrier upgrade. The V/STOL jets would be gradually retired with the *Viraat* around 2010-12.